Multiple Award Contract
Indefinite Delivery/Indefinite Quantity (IDIQ)

STATEMENT OF WORK

COMMERCIAL EARTH OBSERVATION DATA

COMMERCIAL SMALLSAT DATA ACQUISITION (CSDA) PROGRAM

CONTRACT No. TBD

I. INTRODUCTION

In pursuit of NASA's Strategic Objective 1.1 to "Understand The Sun, Earth, Solar System, And Universe - Safeguarding and Improving Life on Earth" (*NASA 2018 Strategic Plan*), the Earth Science Division (ESD) develops unique capabilities in remote sensing to implement a broad suite of space-based Earth observations which, together with many other data, are utilized to advance knowledge of the integrated Earth system, the global atmosphere, oceans (including sea ice), land surfaces, ecosystems, and interactions between all elements, including the impacts of humans. Sustained, simultaneous observations of many geophysical parameters are needed to understand the complexity of the global Earth system.

The quantitative determination of global trends in the Earth's atmosphere, ocean, cryosphere, biosphere, and land surface and interior depends significantly on the availability of multi-instrument/multiplatform data sets, provided at a range of spatial and temporal resolution, which extend to time periods of a decade or longer. The ability to enhance Earth system models and advance predictive capabilities relies on dynamically consistent global observational data sets.

In 2018, the National Aeronautics and Space Administration (NASA) Earth Science Division (ESD) launched the Private-Sector Small Constellation Satellite Data Product Pilot (referred to hereafter as Commercial Smallsat Data Acquisition (CSDA) Program Pilot) with the purpose of identifying and evaluating data that has the potential to support NASA's Earth science research and application activities. The pilot activities ended early 2020, thus transitioning into the Commercial Smallsat Data Acquisition (CSDA) Program. The objective of the CSDA Program is to continue to identify, evaluate and acquire data from commercial sources that support NASA's Earth science research and application activities (https://science.nasa.gov/earth-science). Commercially acquired data may provide a cost-effective means to augment and/or complement the suite of Earth Observations acquired by NASA and other US government agencies and those acquired by international partners and agencies. Emphasis is placed on data acquired by small-satellite constellations, affording the means of complementing NASA acquired data with higher resolutions, increased temporal frequency or other novel capabilities in support of existing Earth science research and application activities.

In this program, NASA desires to procure one or more data sets/information products and associated meta-data (documentation, access routines, and basic characterization information) based on Earth observations derived from Earth-orbiting, small-satellite constellations designed and operated by non-governmental entities, particularly the private sector. NASA intends to provide the purchased data sets to a broad set of separately selected NASA-funded researchers who will be supported to examine and analyze the data set(s) to help determine the utility of the constellation-based products for advancing NASA's science and applications development goals. NASA will collaborate with other U.S. Government agencies and partners during the evaluation phase and/or subsequent purchases.

NASA's Earth Science currently operates an unprecedented number of missions, most of which are past their prime phase and some have operated longer than a decade. Data product developers have matured mission instrument products through refinement of instrument calibration/validation and algorithms. Accordingly, there is increasingly greater focus on research and data production on measurement-based data products, beyond single mission instrument products. In order to create these basic records, a science measurement focus brings together expertise in multiple instrument characterization and calibration, data processing, science-based product generation and distribution, science tools, and interactive relationships with the broader science community.

A broad range of remote sensing techniques and sensors developed by the US private sector community can be implemented in a variety of space flight opportunities. The use of any or all such techniques is permitted under this procurement. Selected data providers will afford NASA, U.S. Government, and its stakeholders an efficient increase in capabilities by building on synergies and complementarity with NASA assets. ESD principles regarding product life cycle planning, and standards and interfaces for interoperability and exchange of data and information will be followed (https://go.usa.gov/xnHCb).

Data products created from multiple instruments, and from "data fusion," remain a challenge, and full utilization of complementary satellite data requires focused research efforts. Here "data fusion" could be low-level fusion, the combining of several sources of raw data to produce new raw data, or data integration, the combining of diverse data sets into a unified data set, which includes all of the data points and time steps. Sophisticated understanding of the contributing instruments' characterization and correct application of the various contributing data must be performed for such records to achieve the expectation that fused data is more informative than the original inputs, while retaining their underlying accuracy.

Emphasis is placed into taking advantage of the opportunities presented by small satellites constellations, providing the means of utilizing a multitude of data sources to form coherent time series, and facilitating the use of consistent clusters of geophysical measurement in the advancement of comprehensive Earth system models.

Specifically, NASA is procuring, for detailed evaluation and assessment, small constellation satellite data products that might augment or complement NASA-collected data. To facilitate standard scientific collaborations, NASA requires End User License Agreements (EULAs) to enable broad levels of dissemination and shareability of the commercial data with the US government agencies and partners. There will be a set of license tiers (i.e., EULA tiers) associated with all contract awards. In addition, all task orders awarded under the IDIQ will identify the EULAs applicable to each task. There may be an initial task order awarded for evaluation purposes to determine the utility of the data for ESD research and applied science applications in a cost-effective way. Upon completion of the evaluation, NASA will determine if any subsequent purchases will be made. Contractors with previously NASA evaluated data via the CSDA Program will not be required to complete the evaluation phase. In case of significant change(s) between the evaluated and currently available data, NASA reserves the right to reevaluate the currently available data (i.e., significant changes to the satellite (s), entirely new commercial product available, etc.). These contractors will be given the opportunity to compete for tasks in the sustained phase. NASA requires the ability to uplift from more restrictive licenses to less restrictive licenses without renegotiation of awards. (See Contract Attachment B for the Government-defined EULAs).

II. SCOPE

Under this Indefinite Delivery/Indefinite Quantity (IDIQ) contract, the vendor shall be responsible for delivery of a comprehensive catalogue of its commercial Earth Observation data products indicating at a minimum: the data sets, associated metadata and ancillary information; data cadence; data latency; area coverage; and data usage policy. The data products, comprising Earth-relevant measurements obtained from commercial small satellite constellations, is incorporated into the master contract as Attachment C. Any revisions or updates to the catalogue shall be coordinated with the Contracting Officer (CO) and Contracting Officer's Representative (COR).

III. REQUIREMENTS

As required, the vendor shall provide access to specific data and other information from its satellites. The vendor data may enhance NASA's assessment of areas in the Earth and applied sciences—e.g. ecosystems, oceanography, geographic and polar sciences—based on both the images and derived products.

The vendor shall deliver Earth-relevant data products obtained from commercial small satellite constellations, associated metadata, and ancillary information. Data, including associated metadata, is needed for the purpose of technical assessment and assimilation into current algorithms and models to determine the viability of these data for existing research and application investigations. The vendor shall provide support on best practices for accessing and using data to address issues that may be encountered in the course of accessing and using the data, as specified in each task order.

All work issued under this contract is subject to the 2003 U.S. Commercial Remote Sensing Policy, as specified in contract article I.16 Supplemental Task Ordering Procedures.

1.0 The vendor's commercial small satellite constellations from which the data is obtained shall be:

a. Comprised of a minimum of at least one Low Earth Oribit (LEO) satellite. The satellite(s) must have been launched, commissioned, and routinely be collecting data.
b. The Low Earth Orbit (LEO) satellite(s) must support consistent global or near-global measurements (e.g., with identical or nearly identical instruments), providing a temporal cadence and spatial coverage that complements the objectives of the NASA Earth Science Division;

OR

c. At least one satellite in higher orbits including medium Earth orbit (MEO), highly eccentric orbit (HEO) or geostationary orbit (GEO), providing spatiotemporal coverage that complements NASA Earth Science objectives;

AND

d. Data from the satellite constellation must cover a period of at least nine (9) continuous months, since 1 June 2016.
e. Satellite data must be nearly globally-acquirable within the revisit constraints of the constellation orbits and observing geometry. For orbits that do not provide such global coverage, a minimum of near-continental-scale (3800 km spatial extent) acquisition capability is required, with a minimum temporal resolution that corresponds to the orbital period (e.g. daily for geostationary orbits).
f. The vendor shall demonstrate that it already has an constellation of small satellites (min. 1 satellite) by:
    - Describing the instrument(s) used in the data acquisition and their sampling characteristics.
    - Describing in detail pre-launch characterization and calibration as well as on-board calibration and stability metrics for each instrument.
    - Describing in detail plans for future satellites.
    - Providing launch date and commissioning information.

2.0 Comprehensive Digital Catalogue

- The vendor shall provide a comprehensive catalogue of its commercial Earth Observation data products indicating at a minimum: the data sets, associated metadata and ancillary information; data cadence; data latency; area coverage; and data usage policy.
- Proposed Earth observation information data products shall be substantially based on measurements obtained from small-satellite constellations (1 or more satellites in orbit at the same time).
- Data products shall have at least continental-scale (3800 km spatial extent) coverage – near-global coverage is preferred (within the constraints of the constellation orbit).
- Data products must cover a period of at least 9 continuous months, beginning no earlier than 1 June 2016 – data products covering longer temporal spans, with minimal gaps, and with high temporal resolution are required.
- Data products that contain per-datum information (Level 0-2, https://go.usa.gov/xnsJK) as opposed to highly processed products (Level 3+) that aggregate observations to a reduced spatiotemporal information content from the measured observation, are required. In the absence of the lowest-level data, detailed documentation must be provided to describe any low-level processing that occurs in the data processing chain.

- Extensive metadata, including details of geophysical algorithms used to process direct observations to geophysical information, any subsequent processing involving combinations of data stream and/or model assimilations and/or space/time averaging and gridding, data characterization/validation including quantitative pre- and post-launch calibration results are preferred. The minimum metadata required are:
    - Variable name
    - Variable unit
    - Unique instrument used for observation
    - Geolocation, including accuracy
    - Variable names for all accompanying geographic and temporal attributes
    - Variable units for all accompanying geographic and temporal attributes
    - If applicable, variable names, description, and units, for any ancillary data included, such as data accuracy, data quality indicators (flags), additional physical parameters, etc.
- Quality and homogeneity of the proposed data products shall be provided by identifying their origin, precision and accuracy estimates.
- Quantification, to the extent possible, of the data uncertainty based on independent data validation shall be provided.

The catalogue, incorporated into the contract as Attachment C, may include one or more of the following (Level definitions are available from https://go.usa.gov/xnsJK):

a. Delivery and distribution of Level 1A and Level 1B products, including instrument calibration information. These products allow the NASA users to develop standard products and allows for the possibility of direct data assimilation into models, depending on the data types.
b. Delivery and distribution of Level 2 products. These products require the delivery of associated Algorithm Theoretical Basis Documents (ATBD).
c. Delivery and distribution of high level (3 and above) products. These products require the delivery of associated Algorithm Theoretical Basis Documents (ATBD) and further documentation regarding the associated uncertainty estimates.

3.0 Professional Services and Support

The vendor may be required to provide the following Professional Services Support, as specified in each task order issued:

- The vendor shall support NASA science evaluation activities related to understanding calibration and geolocation.
- The vendor shall support reformatting data into NASA standard science formats (https://go.usa.gov/xnHCb), including GeoTIFF.
- The vendor shall provide support to address metadata issues for data search and discoverability for long-term use.

- The vendor shall provide, at a minimum, regular metrics of the purchased data and user access (where applicable).
- The vendor shall work with NASA to establish a regular monthly program review for the duration of the task, as specified in each task order.

IV. Data Rights and Policy

Anticipating that at least some of the data sets will be shown to be valuable for advancing NASA science and applications development, NASA has incorporated Government-defined End User License Agreements (EULAs) via Contract Attachment B, for use of these and similar products for evaluation and subsequent longer-term purchase.

NASA has determined a necessity for NASA and other U.S. Government agencies and partners to acquire minimum rights to Data for any and all Data procured under any agreement to support intended Scientific Non-Commercial Use. Scientific Non-Commercial Use means use by Licensed Users of the Data pursuant to a Government initiated, US Government-funded and/or US Government-peer reviewed investigation established through a Government Research Announcement or similar public notice of opportunity, and performed for the sole purpose of conducting experiments, evaluation, research, and/or development, including basic and applied research under a Government Science Program. At a minimum, the U.S. Government and its partners shall have the ability to share and use Data and any Derived Products for Scientific Use including but not limited to inclusion in scientific and technical articles and publishing academic or similar works. The minimum data rights stated above shall apply to all phases of the CSDA Program. Please see Table 1 for the summary of the EULAs intended for the CSDA. The EULAs in their entirety are provided in Contract Attachment B.

Table 1. End User License Agreements (EULAs) Summary

Authorized User Community	Type of EULA		
	Public Release	U.S. Gov Plus	U.S. Gov
U.S. Government defined under Title 5 U.S.C. 101-105	✓	✓	✓
The U.S. Gov Executive Office of the President (EOP), members of Congress, and Congressional staff	✓	✓	✓
State and Local Governments, Territories, and Tribal Authorities within the U.S.	✓	✓	✓
Non-Governmental Organizations and/or Non-Profit Organizations working for the purpose of U.S. Gov	✓	✓	✓
Contractors, subcontractors, partners, and/or grantees supporting the U.S. gov to execute their contracts	✓	✓	✓
Foreign Governments, intergovernmental entities, and International Defense and Coalition Partners for U.S. Gov purposes	✓	✓	
Public, Open Data	✓		

NASA requires the ability to uplift from more restrictive licenses to less restrictive licenses without renegotiation of awards. NASA will notify the vendor when a EULA uplift is required via Task Order Request.

As needed, EULA discussions will be used to inform NASA's policies and approaches for engaging more substantively with private sector small-satellite constellation operators as well as other U.S. Government agencies and partners regarding future data purchases.

As per the *Data and Information Policy* (https://www.earthdata.nasa.gov/engage/open-data-services-and-software/data-and-information-policy), NASA's Earth Science program was established to use the advanced technology of NASA to understand and protect our home planet by using our view from space to study the Earth system and improve prediction of Earth system change. To meet this challenge, NASA promotes the full and open sharing of all data with the research and applications communities, private industry, academia, and the general public. The greater the availability of the data, the more quickly and effectively the user communities can utilize the information to address basic Earth science questions and provide the basis for developing innovative practical applications to benefit the general public.

A common set of carefully crafted data exchange and access principles was created by the Japanese, European and U.S. International Earth Observing System (IEOS) partners during the 1990s and the early years of the 21st century. From these principles, NASA has adopted the Data Policy described at https://www.earthdata.nasa.gov/engage/open-data-services-and-software/data-and-information-policy (in this context the term 'data' includes observation data, metadata, products, information, algorithms, including scientific source code, documentation, models, images, and research results):

The data collected by NASA represent a significant public investment in research. NASA holds these data in a public trust to promote comprehensive, long-term Earth science research. Consequently, NASA developed policy consistent with existing international policies to maximize access to data and to keep user costs as low as possible. These policies apply to all data archived, maintained, distributed or produced by NASA data systems. Non-NASA data are covered by the license arrangement of the organizations that sponsor that dataset.

V. DELIVERABLES

As specified in each task order: The vendor shall make the required data available for access in accordance with the Government-defined EULAs. The vendor shall make data products available from a secure HTTPS location for download by the approved users (i.e. Licensed Users per EULA) along with checksums to ensure integrity of each file. The HTTPS location shall support automated download of data. The vendor shall make data available from the secure HTTPS location for the duration of the award. The vendor shall provide internet access to data with a total bandwidth no less than 2.5GBps. The vendor shall provide data system capabilities for search, discover, and access of the data via (i) user interface and (ii) application programming interface. NASA will maintain a copy of all data products purchased for use by current and future approved researchers. The vendor shall notify the (CO) and (COR) when the package(s) is available for download by NASA. Each task order issued will specify the type of data and EULA required.

VI. SECURITY REQUIREMENTS

For purposes of this IDIQ contract, the vendor will have no logical access to internal NASA IT systems – the only access will be as specified in section V above. Any NASA personnel with authorized user IT accounts must complete NASA IT Security Awareness Training in accordance with the NASA requirements.

VII. GOVERNMENT FURNISHED EQUIPMENT

The government will not supply equipment, software, or access to government facilities for this effort.

VIII. TRAVEL

No travel will be required in the performance of this Contract.

REVISIONS

Revision Number	Date	Brief Summary of Changes
Base	March 2023	N/A
1	January 2024	Update Data and Information Policy Links in Section IV.
2		
3		
4		
5		
6		
7		
8		